Study of Various Network Commands used in Linux and Windows

Exp. no: 1

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Aim :

To study and practice various networking commands used in Linux and Windows for networking configuration, troubleshooting and diagnostics.

Commands :

1.) ARP (Address Resolution protocol)

- Displays and manages the ARP table, which maps Ip addresses to MAC addresses.

```
C:\Windows\system32>arp

Displays and modifies the IP-to-Physical address translation tables used by
address resolution protocol (ARP).

ARP -s inet_addr eth_addr [if_addr]
ARP -d inet_addr [if_addr]
ARP -a [inet_addr] [-N if_addr] [-v]

  -a            Displays current ARP entries by interrogating the current
                protocol data.  If inet_addr is specified, the IP and Physical
                addresses for only the specified computer are displayed.  If
                more than one network interface uses ARP, entries for each ARP
                table are displayed.
  -g            Same as -a.
  -v            Displays current ARP entries in verbose mode.  All invalid
                entries and entries on the loop-back interface will be shown.
  inet_addr     Specifies an internet address.
  -N if_addr    Displays the ARP entries for the network interface specified
                by if_addr.
  -d            Deletes the host specified by inet_addr. inet_addr may be
                wildcarded with * to delete all hosts.
  -s            Adds the host and associates the Internet address inet_addr
                with the Physical address eth_addr.  The Physical address is
                given as 6 hexadecimal bytes separated by hyphens. The entry
                is permanent.
  eth_addr      Specifies a physical address.
  if_addr       If present, this specifies the Internet address of the
                interface whose address translation table should be modified.
                If not present, the first applicable interface will be used.

Example:
  > arp -s 157.55.85.212   00-aa-00-62-c6-09  .... Adds a static entry.
  > arp -a                                    .... Displays the arp table.

C:\Windows\system32>
```

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2.) Hostname

Displays or changes the name of the local computer.

```
C:\Windows\system32>hostname
DESKTOP-G0LVE4J

C:\Windows\system32>_
```

3.) IPConfig (Windows) / ifconfig (Linux)

Displays and configures the Ip address, subnet mask, and default gateway.

```
C:\Windows\system32>ipconfig

Windows IP Configuration


Ethernet adapter Npcap Loopback Adapter:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::b923:c6f7:7122:8700%6
   Autoconfiguration IPv4 Address. . : 169.254.158.128
   Subnet Mask . . . . . . . . . . . : 255.255.0.0
   Default Gateway . . . . . . . . . :

Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::390c:dc1c:3ca7:94bc%3
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   IPv4 Address. . . . . . . . . . . : 172.16.10.86
   Subnet Mask . . . . . . . . . . . : 255.255.252.0
   Default Gateway . . . . . . . . . : 172.16.8.1

C:\Windows\system32>_
```

4.) NBTStat (NetBIOS Statistics)

Displays NetBIOS over TCP/IP protocol statistics and NetBIOS name tables.

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```
C:\Windows\system32>nbtstat

Displays protocol statistics and current TCP/IP connections using NBT
(NetBIOS over TCP/IP).

NBTSTAT [ [-a RemoteName] [-A IP address] [-c] [-n]
        [-r] [-R] [-RR] [-s] [-S] [interval] ]

  -a   (adapter status) Lists the remote machine's name table given its name
  -A   (Adapter status) Lists the remote machine's name table given its
                        IP address.
  -c   (cache)          Lists NBT's cache of remote [machine] names and their IP addresses
  -n   (names)          Lists local NetBIOS names.
  -r   (resolved)       Lists names resolved by broadcast and via WINS
  -R   (Reload)         Purges and reloads the remote cache name table
  -S   (Sessions)       Lists sessions table with the destination IP addresses
  -s   (sessions)       Lists sessions table converting destination IP
                        addresses to computer NETBIOS names.
  -RR  (ReleaseRefresh) Sends Name Release packets to WINS and then, starts Refresh

  RemoteName   Remote host machine name.
  IP address   Dotted decimal representation of the IP address.
  interval     Redisplays selected statistics, pausing interval seconds
               between each display. Press Ctrl+C to stop redisplaying
               statistics.


C:\Windows\system32>
```

5.) Netstat

Displays active network connections, routing tables and listening ports.

```
C:\Windows\system32>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    127.0.0.1:49675        DESKTOP-G0LVE4J:49676  ESTABLISHED
  TCP    127.0.0.1:49676        DESKTOP-G0LVE4J:49675  ESTABLISHED
  TCP    127.0.0.1:49677        DESKTOP-G0LVE4J:49678  ESTABLISHED
  TCP    127.0.0.1:49678        DESKTOP-G0LVE4J:49677  ESTABLISHED
  TCP    172.16.10.86:13133     4.213.25.242:https     ESTABLISHED
  TCP    172.16.10.86:13199     lcbomp-in-f188:5228    ESTABLISHED
  TCP    172.16.10.86:13258     a23-202-229-23:https   CLOSE_WAIT
  TCP    172.16.10.86:13259     150.171.110.82:https   CLOSE_WAIT
  TCP    172.16.10.86:13260     131.253.33.254:https   CLOSE_WAIT
  TCP    172.16.10.86:13261     135.233.41.222:https   CLOSE_WAIT
  TCP    172.16.10.86:13262     204.79.197.222:https   CLOSE_WAIT
  TCP    172.16.10.86:13270     72.154.187.226:https   CLOSE_WAIT
  TCP    172.16.10.86:13272     20.100.121.19:https    CLOSE_WAIT
  TCP    172.16.10.86:13273     a-0003:http            CLOSE_WAIT
  TCP    172.16.10.86:13278     13.107.42.254:https    CLOSE_WAIT
  TCP    172.16.10.86:13279     52.108.9.254:https     CLOSE_WAIT
  TCP    172.16.10.86:13281     172.202.65.254:https   CLOSE_WAIT
  TCP    172.16.10.86:13293     a23-211-142-112:https  TIME_WAIT
  TCP    172.16.10.86:13294     a23-211-142-112:https  TIME_WAIT
  TCP    172.16.10.86:13295     a23-211-142-112:https  TIME_WAIT
  TCP    172.16.10.86:13297     bkk02s04-in-f14:https  ESTABLISHED
  TCP    172.16.10.86:13298     4.213.25.242:https     ESTABLISHED
  TCP    [fe80::390c:dc1c:3ca7:94bc%3]:1521  DESKTOP-G0LVE4J:49679  ESTABLISHED
  TCP    [fe80::390c:dc1c:3ca7:94bc%3]:49679  DESKTOP-G0LVE4J:1521  ESTABLISHED

C:\Windows\system32>
```

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6.) NsLookup

Queries DNS servers to find the Ip address or domain name of a host.

```
C:\Windows\system32>nslookup www.google.com
Server:  UnKnown
Address:  172.16.8.1

Non-authoritative answer:
Name:    www.google.com
Addresses:  2001:4860:4827:7700::
          2001:4860:482b:7700::
          2001:4860:4828:7700::
          2001:4860:482a:7700::
          2001:4860:4826:7700::
          2001:4860:4829:7700::
          2001:4860:482d:7700::
          2001:4860:482c:7700::
          142.251.156.119
          142.251.155.119
          142.251.151.119
          142.251.150.119
          142.251.157.119
          142.251.152.119
          142.251.154.119
          142.251.153.119


C:\Windows\system32>
```

7.) Pathping

combines the features of ping and Tracecuts to identify packet loss and network latency.

```
C:\Windows\system32>pathping

Usage: pathping [-g host-list] [-h maximum_hops] [-i address] [-n]
                [-p period] [-q num_queries] [-w timeout]
                [-4] [-6] target_name

Options:
    -g host-list      Loose source route along host-list.
    -h maximum_hops   Maximum number of hops to search for target.
    -i address        Use the specified source address.
    -n                Do not resolve addresses to hostnames.
    -p period         Wait period milliseconds between pings.
    -q num_queries    Number of queries per hop.
    -w timeout        Wait timeout milliseconds for each reply.
    -4                Force using IPv4.
    -6                Force using IPv6.

C:\Windows\system32>_
```

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8) ping (packet Internet Groper)

Tests network connectivity between two devices by sending ICMP echo requests.

```
C:\Windows\system32>ping

Usage: ping [-t] [-a] [-n count] [-l size] [-f] [-i TTL] [-v TOS]
            [-r count] [-s count] [[-j host-list] | [-k host-list]]
            [-w timeout] [-R] [-S srcaddr] [-c compartment] [-p]
            [-4] [-6] target_name

Options:
    -t             Ping the specified host until stopped.
                   To see statistics and continue - type Control-Break;
                   To stop - type Control-C.
    -a             Resolve addresses to hostnames.
    -n count       Number of echo requests to send.
    -l size        Send buffer size.
    -f             Set Don't Fragment flag in packet (IPv4-only).
    -i TTL         Time To Live.
    -v TOS         Type Of Service (IPv4-only. This setting has been deprecated
                   and has no effect on the type of service field in the IP
                   Header).
    -r count       Record route for count hops (IPv4-only).
    -s count       Timestamp for count hops (IPv4-only).
    -j host-list   Loose source route along host-list (IPv4-only).
    -k host-list   Strict source route along host-list (IPv4-only).
    -w timeout     Timeout in milliseconds to wait for each reply.
    -R             Use routing header to test reverse route also (IPv6-only).
                   Per RFC 5095 the use of this routing header has been
                   deprecated. Some systems may drop echo requests if
                   this header is used.
    -S srcaddr     Source address to use.
    -c compartment Routing compartment identifier.
    -p             Ping a Hyper-V Network Virtualization provider address.
    -4             Force using IPv4.
    -6             Force using IPv6.
```

9) Route

Displays or modifies the IP routing table used for packet forwarding.

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```
C:\Windows\system32>route

Manipulates network routing tables.

ROUTE [-f] [-p] [-4|-6] command [destination]
                  [MASK netmask]  [gateway] [METRIC metric]  [IF interface]

  -f            Clears the routing tables of all gateway entries.  If this is
                used in conjunction with one of the commands, the tables are
                cleared prior to running the command.

  -p            When used with the ADD command, makes a route persistent across
                boots of the system. By default, routes are not preserved
                when the system is restarted. Ignored for all other commands,
                which always affect the appropriate persistent routes.

  -4            Force using IPv4.

  -6            Force using IPv6.

  command       One of these:
                  PRINT     Prints  a route
                  ADD       Adds    a route
                  DELETE    Deletes a route
                  CHANGE    Modifies an existing route
  destination   Specifies the host.
  MASK          Specifies that the next parameter is the 'netmask' value.
  netmask       Specifies a subnet mask value for this route entry.
                If not specified, it defaults to 255.255.255.255.
  gateway       Specifies gateway.
  interface     the interface number for the specified route.
  METRIC        specifies the metric, ie. cost for the destination.

All symbolic names used for destination are looked up in the network database
file NETWORKS. The symbolic names for gateway are looked up in the host name
database file HOSTS.

If the command is PRINT or DELETE. Destination or gateway can be a wildcard,
(wildcard is specified as a star '*'), or the gateway argument may be omitted.

If Dest contains a * or ?, it is treated as a shell pattern, and only
matching destination routes are printed. The '*' matches any string,
and '?' matches any one char. Examples: 157.*.1, 157.*, 127.*, *224*.

Pattern match is only allowed in PRINT command.
Diagnostic Notes:
    Invalid MASK generates an error, that is when (DEST & MASK) != DEST.
    Example> route ADD 157.0.0.0 MASK 155.0.0.0 157.55.80.1 IF 1
             The route addition failed: The specified mask parameter is invalid. (Destination & Mask) != Destination.

Examples:

    > route PRINT
    > route PRINT -4
    > route PRINT -6
    > route PRINT 157*            .... Only prints those matching 157*

    > route ADD 157.0.0.0 MASK 255.0.0.0  157.55.80.1 METRIC 3 IF 2
              destination^      ^mask      ^gateway     metric^    ^
                                                        Interface^
      If IF is not given, it tries to find the best interface for a given
      gateway.
    > route ADD 3ffe::/32 3ffe::1

    > route CHANGE 157.0.0.0 MASK 255.0.0.0 157.55.80.5 METRIC 2 IF 2

      CHANGE is used to modify gateway and/or metric only.

    > route DELETE 157.0.0.0
    > route DELETE 3ffe::/32

C:\Windows\system32>_
```

Result :

The networking commands were successfully executed and studied. Their outputs were observed and used to understand network config, connectivity testing, DNS resolution, routing and troubleshooting.

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Some important Linux networking Commands.

1. ip

The ip command is one of the basic commands every administrator will need in daily work, from setting up new systems and assigning IP's to troubleshooting existing systems.

```
[root@localhost ~]# ip
Usage: ip [ OPTIONS ] OBJECT { COMMAND | help }
       ip [ -force ] -batch filename
where  OBJECT := { address | addrlabel | fou | help | ila | ioam | l2tp | link |
                   macsec | maddress | monitor | mptcp | mroute | mrule |
                   neighbor | neighbour | netconf | netns | nexthop | ntable |
                   ntbl | route | rule | sr | stats | tap | tcpmetrics |
                   token | tunnel | tuntap | vrf | xfrm }
       OPTIONS := { -V[ersion] | -s[tatistics] | -d[etails] | -r[esolve] |
                    -h[uman-readable] | -iec | -j[son] | -p[retty] |
                    -f[amily] { inet | inet6 | mpls | bridge | link } |
                    -4 | -6 | -M | -B | -0 |
                    -l[oops] { maximum-addr-flush-attempts } | -echo | -br[ief] |
                    -o[neline] | -t[imestamp] | -ts[hort] | -b[atch] [filename] |
                    -rc[vbuf] [size] | -n[etns] name | -N[umeric] | -a[ll] |
                    -c[olor]}
[root@localhost ~]#
```

a.) ip address show

```
[root@localhost ~]# ip address show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:cb:73:6a brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
       valid_lft 83740sec preferred_lft 83740sec
    inet6 fe80::a00:27ff:fecb:736a/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
[root@localhost ~]#
```

b.) ip address add 192.168.1.254/24 dev enpos3

c.) ip address del 192.168.1.254/24 dev enpos3

d.) ip link set enpos3 up

e.) ip link set enpos3 down

d.) ip link set enpos3 promisc on

e.) ip route get 10.10.1.4

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```
[root@localhost ~]# ip address add 192.168.1.254/24 dev enp0s3
[root@localhost ~]# ip address del 192.168.1.254/24 dev enp0s3
[root@localhost ~]# ip link set enp0s3 up
[root@localhost ~]# ip link set enp0s3 down
[root@localhost ~]# ip link set enp0s3 promisc on

[deepak3@localhost ~]$ ip route get 10.10.1.4
10.10.1.4 via 10.0.2.2 dev enp0s3 src 10.0.2.15 uid 1000
    cache
```

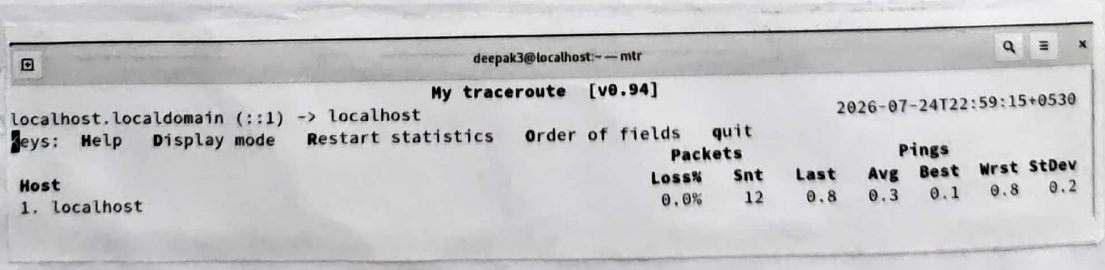
2.) ifconfig

The ifconfig command was / is a staple in many sysadmin's tool belt for configuring and troubleshooting networks.

```
[root@localhost ~]# ifconfig
lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 106  bytes 7915 (7.7 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 106  bytes 7915 (7.7 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

3.) mtr

MTR (Matt's traceroute) is a program with a command-line interface that serves as a network diagnostic and troubleshooting tool.



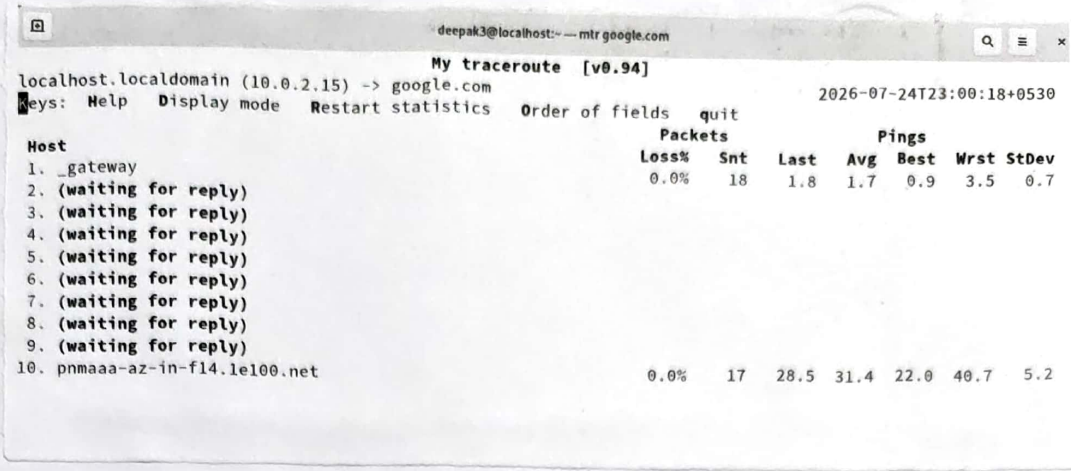
a) mtr google.com

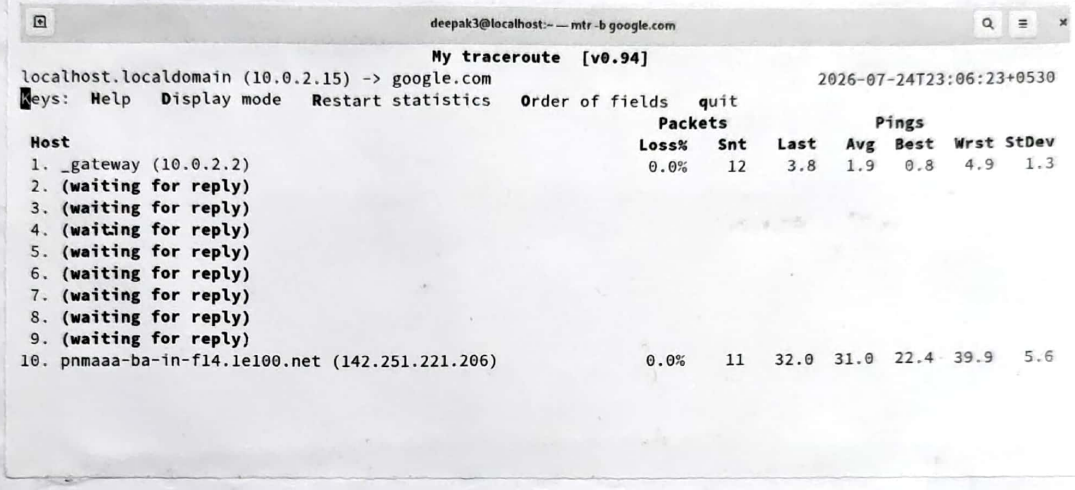
b) mtr -g google.com

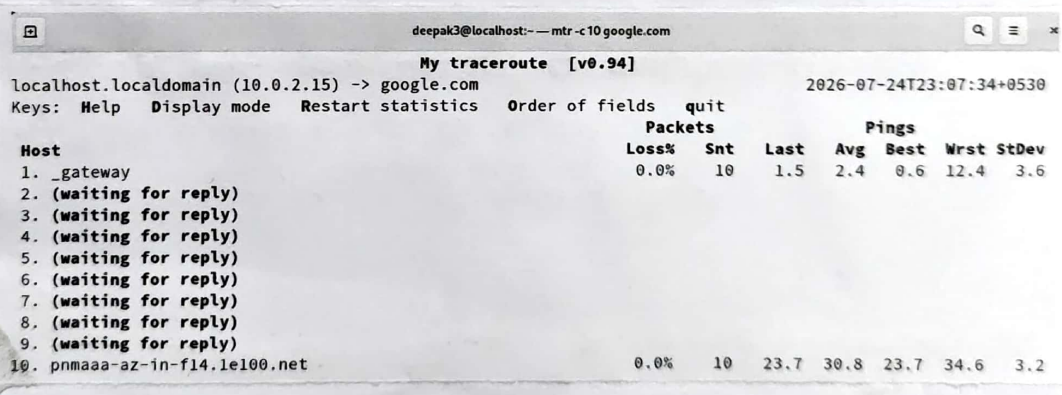
c) mtr -b google.com

d) mtr -c 10 google.com

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4.) tcpdump

The tcpdump command is designed for capturing and displaying packets.

a.) dnf install -y tcpdump

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```
[root@localhost ~]# tcpdump
dropped privs to tcpdump
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on any, link-type LINUX_SLL (Linux cooked v1), snapshot length 262144 bytes

^C
0 packets captured
0 packets received by filter
0 packets dropped by kernel
[root@localhost ~]# su -
[root@localhost ~]# dnf install -y tcpdump
CentOS Stream 9 - BaseOS                                        0.0  B/s |   0  B     00:00
Errors during downloading metadata for repository 'baseos':
  - Curl error (6): Couldn't resolve host name for https://mirrors.centos.org/metalink?rep
o=centos-baseos-9-stream&arch=x86_64&protocol=https,http [Could not resolve host: mirrors.
centos.org]
Error: Failed to download metadata for repo 'baseos': Cannot prepare internal mirrorlist:
Curl error (6): Couldn't resolve host name for https://mirrors.centos.org/metalink?repo=ce
ntos-baseos-9-stream&arch=x86_64&protocol=https,http [Could not resolve host: mirrors.cent
os.org]
[root@localhost ~]#
```

b) tcpdump -D

```
[root@localhost ~]# tcpdump -D
1.any (Pseudo-device that captures on all interfaces) [Up, Running]
2.lo [Up, Running, Loopback]
3.bluetooth-monitor (Bluetooth Linux Monitor) [Wireless]
4.usbmon2 (Raw USB traffic, bus number 2)
5.usbmon1 (Raw USB traffic, bus number 1)
6.usbmon0 (Raw USB traffic, all USB buses) [none]
7.nflog (Linux netfilter log (NFLOG) interface) [none]
8.nfqueue (Linux netfilter queue (NFQUEUE) interface) [none]
9.enp0s3 [none, Disconnected]
```

c) (i) capture traffic to and from one host
(ii) capture traffic to and from a network.
(iii) capture traffic to and from port numbers.

```
[root@localhost ~]# tcpdump -i enp0s3 -c 10 'host 8.8.8.8'
dropped privs to tcpdump
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on enp0s3, link-type EN10MB (Ethernet), snapshot length 262144 bytes
^C
0 packets captured
0 packets received by filter
0 packets dropped by kernel
[root@localhost ~]# tcpdump -i enp0s3 net 10.1.0.0/24
dropped privs to tcpdump
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on enp0s3, link-type EN10MB (Ethernet), snapshot length 262144 bytes
-v[v]
^C
0 packets captured
0 packets received by filter
0 packets dropped by kernel
```

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5.) ping

ping is a tool that verify IP-level connectivity to another TCP/IP computers by sending Internet Control Message protocol (ICMP) Echo Request messages. a.) ping google.com

```
[root@localhost ~]# ping google.com
PING google.com (142.251.221.206) 56(84) bytes of data.
64 bytes from pnmaaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=1 ttl=117 time=27.0 m
s
64 bytes from pnmaaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=2 ttl=117 time=30.6 m
s
64 bytes from pnmaaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=3 ttl=117 time=36.8 m
s
64 bytes from pnmaaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=4 ttl=117 time=48.0 m
s
64 bytes from pnmaaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=5 ttl=117 time=38.9 m
s
64 bytes from pnmaaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=6 ttl=117 time=36.7 m
s
64 bytes from pnmaaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=7 ttl=117 time=40.7 m
s
64 bytes from pnmaaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=8 ttl=117 time=27.3 m
s
64 bytes from pnmaaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=9 ttl=117 time=24.7 m
s
64 bytes from pnmaaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=10 ttl=117 time=31.6
ms
64 bytes from pnmaaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=11 ttl=117 time=49.9
ms
```

b.) ping -c 10 google.com.

```
[root@localhost ~]# ping -c 10 google.com
PING google.com (142.251.222.142) 56(84) bytes of data.
64 bytes from pnmaaa-az-in-f14.1e100.net (142.251.222.142): icmp_seq=1 ttl=117 time=29.2 ms
64 bytes from pnmaaa-az-in-f14.1e100.net (142.251.222.142): icmp_seq=2 ttl=117 time=26.4 ms
64 bytes from pnmaaa-az-in-f14.1e100.net (142.251.222.142): icmp_seq=3 ttl=117 time=54.3 ms
64 bytes from pnmaaa-az-in-f14.1e100.net (142.251.222.142): icmp_seq=4 ttl=117 time=30.7 ms
64 bytes from pnmaaa-az-in-f14.1e100.net (142.251.222.142): icmp_seq=5 ttl=117 time=32.0 ms
64 bytes from pnmaaa-az-in-f14.1e100.net (142.251.222.142): icmp_seq=6 ttl=117 time=47.6 ms
64 bytes from pnmaaa-az-in-f14.1e100.net (142.251.222.142): icmp_seq=7 ttl=117 time=45.6 ms
64 bytes from pnmaaa-az-in-f14.1e100.net (142.251.222.142): icmp_seq=8 ttl=117 time=45.0 ms
64 bytes from pnmaaa-az-in-f14.1e100.net (142.251.222.142): icmp_seq=9 ttl=117 time=41.5 ms
64 bytes from pnmaaa-az-in-f14.1e100.net (142.251.222.142): icmp_seq=10 ttl=117 time=28.2 ms

--- google.com ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9019ms
rtt min/avg/max/mdev = 26.429/38.064/54.310/9.343 ms
```

Configuring an Ethernet connection by using nmcli.

If you connect a host to the network over ethernet, you can manage the connection's

settings on the command line by using the nmcli utility.

procedure :-

1) List the Network Manager connection profiles

```
[root@localhost ~]# nmcli connection show
NAME    UUID                                  TYPE      DEVICE
enp0s3  5c98baec-516a-3019-bcd6-c952b809ade2  ethernet  enp0s3
lo      1b21ce29-134f-400b-b37c-5521f6847f65  loopback  lo
[root@localhost ~]#
```

2) modify

3) Rename the Connection profile

4) Display the current settings of the connection profile.

```
[root@localhost ~]# nmcli connection add con-name enp0s3 ifname enp0s3 type ethernet
Warning: There is another connection with the name 'enp0s3'. Reference the connection by i
ts uuid '29ed6a04-c4f7-4e59-afd0-1027fbbab918'
Connection 'enp0s3' (29ed6a04-c4f7-4e59-afd0-1027fbbab918) successfully added.
[root@localhost ~]#
```

```
[root@localhost ~]# nmcli connection modify "enp0s3"
Warning: There are 2 other connections with the name 'enp0s3'. Reference the connection by
 its uuid '5c98baec-516a-3019-bcd6-c952b809ade2'
[root@localhost ~]# nmcli connection show
NAME    UUID                                  TYPE      DEVICE
enp0s3  5c98baec-516a-3019-bcd6-c952b809ade2  ethernet  enp0s3
lo      1b21ce29-134f-400b-b37c-5521f6847f65  loopback  lo
enp0s3  29ed6a04-c4f7-4e59-afd0-1027fbbab918  ethernet  --
enp0s3  04bba68c-9e05-4e71-8a76-a552d68783e4  ethernet  --
[root@localhost ~]#
```

5) Configure the Ipv4 settings

6) Configure the Ipv6 settings

7) Activate the profile.

```
[root@localhost ~]# nmcli connection modify "enp0s3" ipv4.method auto
[root@localhost ~]# nmcli connection modify "enp0s3" ipv4.method manual ipv4.addresses 192
.0.2.1/24 ipv4.gateway 192.0.2.254 ipv4.dns 192.0.2.200 ipv4.dns-search example.com
[root@localhost ~]# nmcli connection modify "enp0s3" ipv6.method auto
[root@localhost ~]# nmcli connection modify "enp0s3" ipv6.method manual ipv6.addresses 200
1:db8:1::fffe/64 ipv6.gateway 2001:db8:1::fffe ipv6.dns 2001:db8:1::ffbb ipv6.dns-search e
xample.com
[root@localhost ~]# nmcli connection up enp0s3
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/Acti
veConnection/3)
[root@localhost ~]#
```

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Verification

1.) ip address show enp0s3

```
[root@localhost ~]# ip address show enp0s3
2: enp0s3: <BROADCAST,MULTICAST,PROMISC,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP grou
p default qlen 1000
    link/ether 08:00:27:cb:73:6a brd ff:ff:ff:ff:ff:ff
    inet 192.0.2.1/24 brd 192.0.2.255 scope global noprefixroute enp0s3
       valid_lft forever preferred_lft forever
    inet6 2001:db8:1::fffe/64 scope global noprefixroute
       valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fecb:736a/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
[root@localhost ~]#
```

2.) ip route show default

3.) ip -6 route show default

4.) cat /etc/resolv.conf

```
[root@localhost ~]# ip route show default
default via 192.0.2.254 dev enp0s3 proto static metric 100
[root@localhost ~]# ip -6 route show default
[root@localhost ~]# cat /etc/resolv.conf
# Generated by NetworkManager
search example.com
nameserver 192.0.2.200
nameserver 2001:db8:1::ffbb
```

5.) ping <host-name -or -Ip-address>

```
[root@localhost ~]# ping 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
From 192.0.2.1 icmp_seq=1 Destination Host Unreachable
From 192.0.2.1 icmp_seq=2 Destination Host Unreachable
From 192.0.2.1 icmp_seq=3 Destination Host Unreachable
From 192.0.2.1 icmp_seq=4 Destination Host Unreachable
From 192.0.2.1 icmp_seq=5 Destination Host Unreachable
From 192.0.2.1 icmp_seq=6 Destination Host Unreachable
From 192.0.2.1 icmp_seq=7 Destination Host Unreachable
From 192.0.2.1 icmp_seq=8 Destination Host Unreachable
From 192.0.2.1 icmp_seq=9 Destination Host Unreachable
From 192.0.2.1 icmp_seq=10 Destination Host Unreachable
From 192.0.2.1 icmp_seq=11 Destination Host Unreachable
From 192.0.2.1 icmp_seq=12 Destination Host Unreachable
From 192.0.2.1 icmp_seq=13 Destination Host Unreachable
From 192.0.2.1 icmp_seq=14 Destination Host Unreachable
From 192.0.2.1 icmp_seq=15 Destination Host Unreachable
From 192.0.2.1 icmp_seq=16 Destination Host Unreachable
From 192.0.2.1 icmp_seq=17 Destination Host Unreachable
```

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Troubleshooting

→ verify that the network cable is plugged in to the host and a switch.

→ Check whether the link failure exists only on this host or also on other hosts connected to the same switch.

Result :

The networking commands for windows and Linux were successfully executed and studied. Their outputs were observed and used to understand network config, connectivity testing, routing and DNS resolution, troubleshooting.

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